

TEAM CYBERGUARD

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Setup Guide

Find Evil! — Autonomous DFIR Triage Agent

This guide explains how to reach the live demo, set the project up locally on a SIFT Workstation, and use the application to run an investigation.

Find Evil! collects forensic evidence read-only, reasons over it like a senior analyst, self-corrects, and produces a severity-ranked, fully traceable report. Disk images are processed with The Sleuth Kit; network captures are decoded with tshark; an AI agent performs the analysis.

OVERVIEW

Technology Stack

Layer	Component
Forensic tooling	SANS SIFT Workstation (The Sleuth Kit, tshark, ...)
Collection	collect.sh (disk images) · collect_pcap.sh (network captures)
Agent	Claude via the claude CLI — read-only dontAsk mode
Orchestration	Python agent that streams the reasoning live
Interface	Streamlit web app (login, dashboard, live analysis)

LIVE DEMO · RECOMMENDED

Access the Deployed App

The application is already deployed — no setup is required. Sign in with the demo credentials (see Local Setup § 5), then open the Analyze tab to run an investigation.

<http://32.192.194.180:8501/>

LOCAL SETUP

SIFT Workstation

1. Prerequisites

- Ubuntu 22.04+ (or the SANS SIFT Workstation)
- Python 3.11+ (3.10 also works — TOML parsing has a fallback)
- The claude CLI (Claude Code) and a valid Anthropic API key
- SIFT forensic tools available on PATH

```
fls -V          # The Sleuth Kit (disk images)
tshark -v       # Wireshark CLI (network captures)
```

2. Clone the repository

```
git clone https://github.com/sanika1323/find-evil-hackathon.git
cd find-evil-hackathon
```

3. Create the Python environment

```
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

4. Configure the Claude API key

```
export ANTHROPIC_API_KEY="YOUR_API_KEY"
```

5. Configure login credentials

Authentication is read from creds.toml (plaintext, or sha256:<hash>). The default demo user is analyst / findevil2024.

```
cp creds.toml.example creds.toml
```

6. Run the application

```
streamlit run app.py --server.address 0.0.0.0
# then open http://localhost:8501
```

USING THE APP

Workspace & Workflow

Sign in on the landing screen, then navigate using the left sidebar. Use Log out at the bottom of the sidebar to end the session.

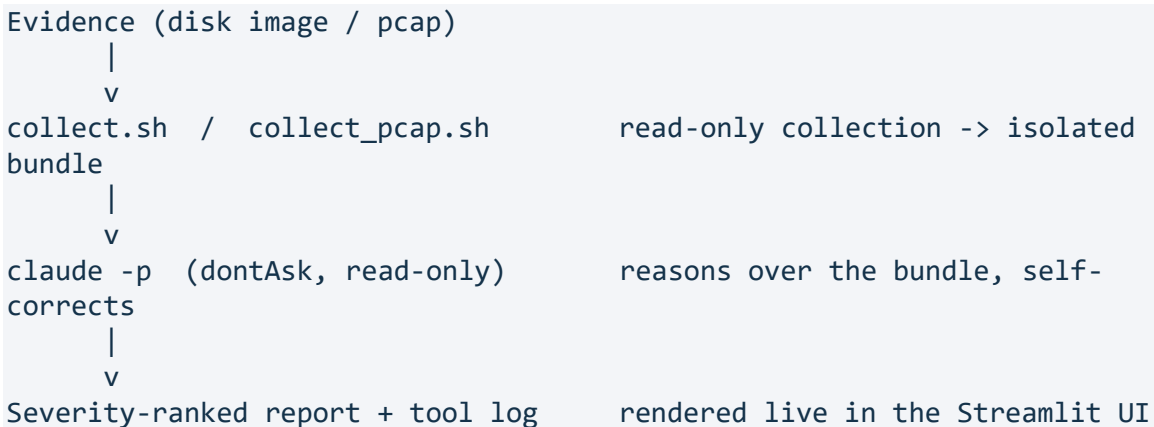
Tab	Purpose
Dashboard	Product overview, how-it-works, and the latest-run summary
Evidence	Register evidence by path; it is hashed (SHA-256 / MD5) with a chain-of-custody record
Analyze	Choose the evidence type, run the agent, and watch its reasoning stream live
Findings	Severity breakdown, confidence labels, IOC graph, event timeline, downloadable report
Logs	Full traceable tool-call log, accuracy self-assessment, and integrity check

Running an investigation

- Open the Analyze tab.
- Select the Evidence type — Disk image or Network capture — which filters the dropdown.
- Pick the evidence (or paste a full path) and click Run triage agent.
- Watch the live agent stream: collection → each tool call → reasoning → severity-ranked report.

HOW IT WORKS

Collection → Reasoning → Report



- Disk images (.E01 .dd .raw .img .001) — collected with The Sleuth Kit (partition offset, filesystem metadata, indicators, prefetch, Recycle Bin records).
- Network captures (.pcap .pcapng .cap) — decoded with tshark (capture metadata, protocol hierarchy, endpoints, DNS / HTTP / SMTP, credentials, host identities).
- Findings are ranked Critical / High / Medium / Low and labelled CONFIRMED / LIKELY / UNVERIFIED.

Case independence: each run is hermetic — prior evidence bundles are moved out of the working directory before collection, and the agent reports only on the current bundle. Findings never carry over between cases.

INTEGRITY

Evidence Integrity

- Original evidence is never modified — it is opened read-only.
- The agent runs in dontAsk mode with only read-only tools allow-listed; destructive commands (dd, mount, rm, mkfs, tee, ...) are deny-listed.
- A SHA-256 baseline is recorded at intake and re-verified after analysis.
- Every tool call is logged and downloadable — any finding traces back to the exact command that produced it.

OUTPUT

What the Report Contains

- Severity classification with explicit C1 / H1 / M1 / L1 IDs.
- A confidence label per finding (CONFIRMED / LIKELY / UNVERIFIED).
- An evidence reference for every finding (bundle file + inode / source IP).
- Registered owner / primary user (or suspect host) and the likely modus operandi.
- An analyst-style summary, recommended next steps, and a full tool-execution log.

SUPPORT

Troubleshooting

Check	Command
Streamlit installed	<code>streamlit --version</code>
Python version	<code>python3 --version</code>
Sleuth Kit (disk)	<code>fls -V</code>
tshark (network)	<code>tshark -v</code>
Claude CLI	<code>claude --version</code>
API key set	<code>echo \$ANTHROPIC_API_KEY</code>

- Login fails — confirm creds.toml exists and the username / password match.
- Network capture shows no decoded data — install tshark: `sudo apt-get install -y tshark`.
- “Command not found” for forensic tools — ensure SIFT tooling is installed and on PATH.